Topic 2 — Climate of India

Covers syllabus (click to expand)

Climate Basics: Climate of India | Factors Affecting Climate | Temperature Distribution | Effect of Altitude | Heat Budget | Temperature Inversion | Pressure & Wind System | Structure of Atmosphere | Pressure Belts | Planetary Winds | ITCZ **Monsoon System:** Monsoon Mechanism | Southwest Monsoon | Northeast Monsoon | Retreating Monsoon (लौटता मानसून) | Indian Seasons | Western Disturbances **Winds, Storms & Cyclones:** Local Winds | Local Storms | Katabatic (अवरोही पवन) Winds | Anabatic (आरोही) Winds | Loo (लू) | Mango Showers | Blossom Showers | Kalbaisakhi / Norwesters | Cyclones | Cyclone Naming System | Cyclone Basins (Bay of Bengal & Arabian Sea) | Thunderstorms **Classification & Phenomena:** Köppen (कोपेन) Climate Classification (India) | Thornthwaite (थॉर्नथ्वेट) Climate Classification | Jet Stream | Western Jet | Tropical Easterly Jet | ENSO | El Niño (अल नीनो) | La Niña (ला नीना) | Indian Ocean Dipole (IOD)

Sources baked in: NCERT Geography Class 11 (Ch 4–5), Class 12 climatology, UPPCS Prelims PYQs 2018–2025 **Weight:** ★★★ — monsoon traps, Western Disturbances, jets, El Niño, local winds, cyclone names **Last verified:** August 2026 **Current Affairs:** IMD 2026 SW monsoon **below normal (~90% LPA)**; ENSO shifting toward **El Niño**; IOD **neutral** (no +IOD offset)

Current Affairs (this topic)

Year	Fact	Why asked	Source
2026	IMD SW monsoon (दक्षिण-पश्चिम) **below normal (~90% of LPA)**	Seasonal rainfall category	IMD LRF Apr/May 2026
2026	ENSO: **neutral** → **El Niño** during Jun–Sep 2026	Weak-monsoon tendency	IMD / MMCFS
2026	**IOD neutral** (no +IOD (धनात्मक) offset of El Niño)	Contrast with 2019 +IOD rescue	IMD Jul 2026
2026	Onset over Kerala **4 June** (slightly after normal 1 June)	Onset date trap	USDA / IMD advance
Ongoing	**Mission Mausam** (MoES/IMD) — improved monsoon & severe-weather forecast	Scheme/institution CA	IMD–MHA

El Niño **tendency** ≠ guaranteed all-India drought (सूखा) — still the 2026 -relevant pairing.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q96

According to Koppen's climatic classification, the (Amw) climate is found predominantly in which part of India?

A. North-Western India B. Western coastal region C. Thar (थार) desert D. Northern plain

Show answer

Logic: A = tropical, w = dry winter — western coastal strip. **Ans: B.** Amw marks the western coastal strip (and parts of the north-east). Thar is BWhw; northern plain largely Cwg.

Q. UKPCS Prelims 2025, Q98

Which local wind affects some parts of Kerala and Tamil Nadu (नाडु) during the pre-monsoon season?

A. Loo B. Mango Shower C. Kal Baisakhi (बैसाखी) D. Mistral

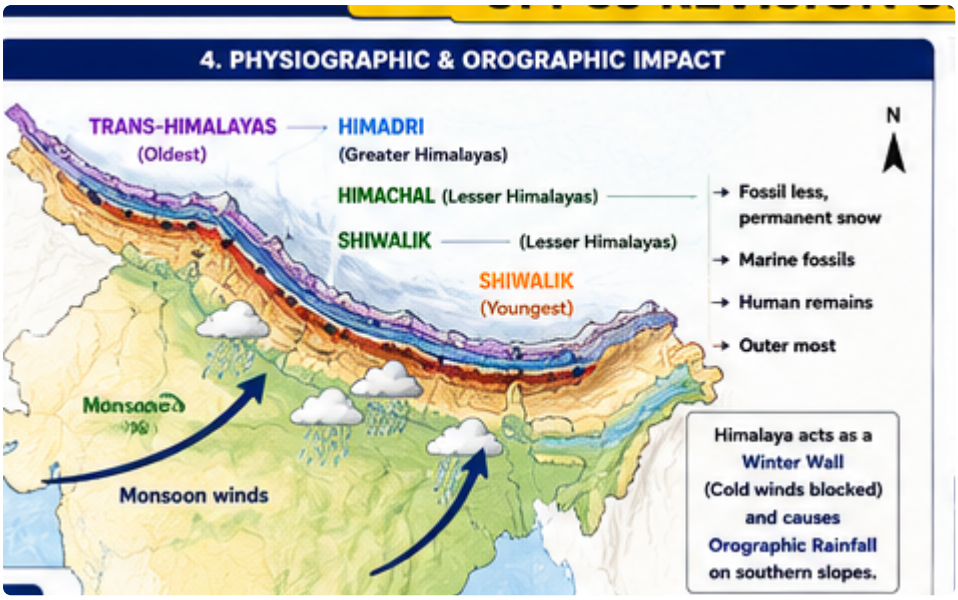
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Logic: Loo and Kal Baisakhi are northern — southern pre-monsoon is mango shower. **Ans: B.** Mango showers help ripen mangoes in Kerala and coastal Tamil Nadu.

2.1 Climate Basics

Climate is the long-term weather pattern of a place. India's climate is **tropical monsoon** because winds reverse with the seasons. Latitude, the Himalaya, the sea, relief, pressure systems, jets, and ENSO/IOD all shape that pattern.

Factor	What it does in India
Latitude	Most of India is tropical / subtropical; the far north is cooler
Himalaya	Winter **wall** against Central Asian cold; summer **orographic** rain on southern slopes
Distance from sea	Coasts stay moderate (नरम दल); NW plains and interior Deccan have **continental** extremes
Altitude	Temperature falls about **6.5 °C per 1000 m** (lapse rate)
Relief / orography	Western Ghats → windward wet, leeward Deccan dry; Meghalaya (मेघालय) scarp → extreme rain
Pressure & ITCZ	Summer thermal **low** + north-shifted ITCZ pull the south-west monsoon
Jet streams	Winter STWJ (पश्चिमी जेट) steers Western Disturbances; summer TEJ (उष्णकटिबंधीय पूर्वी जेट) supports SW monsoon
ENSO / IOD	Year-to-year monsoon strength (El Niño weakens; +IOD helps)

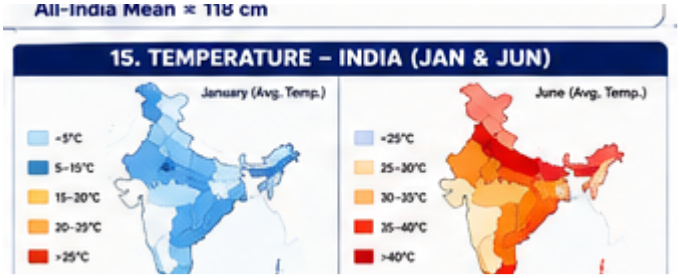


- Latitude places most of India in the tropical and subtropical belts. The Himalaya adds a cold-temperate belt in the north.
- The Himalaya blocks cold Central Asian winds in winter. It also forces orographic rain on the southern slopes in the monsoon.
- Coasts stay moderate because of the sea. The interior Deccan and north-west India have continental extremes.
- The Western Ghats make the Konkan–Malabar windward slope wet. The Deccan to their east is in rain shadow.
- Jet streams, ENSO, and IOD change monsoon strength from year to year. Those mechanisms are taught in full below.

Temperature and altitude

Temperature normally falls with height because the ground heats the lowest air. Higher up, air is thinner and farther from that heat source. The normal lapse rate is about **6.5 °C per 1000 m**.

Station	January (approx.)	Control
Drass / Ladakh (लद्दाख)	−20 to −40 °C	Altitude plus continentality
Agra (आगरा)	about **16 °C**	Ganga plain —
Darjeeling (about 2000 m)	about **4 °C**	Same latitude as Agra; **lapse rate**
Mumbai / Chennai	24–26 °C	Maritime
Jaisalmer	12–15 °C in January; 40–45 °C in June	Highest **diurnal** range among usual stations



- January is coldest in the north-west and the Himalaya. January is warmest in the far south.
- June is hottest in north-west Rajasthan and Gujarat.
- **Diurnal range** means day temperature minus night temperature. Deserts have a large range. Coasts have a small range.
- The **January 18 °C isotherm** is the usual key fact that divides India into tropical (south of it) and subtropical (north of it) zones.
- Same-latitude pairs differ mainly by altitude: **Agra–Darjeeling** and **Amritsar (अमृतसर)–Shimla**.

Climate contrast	Main control
Chennai warmer than Kolkata (annual)	**Latitude** (Chennai farther south)
Himalayan snowfall / hill stations cooler	**Altitude**
Rainfall falls West Bengal → Punjab	**Distance from the Bay / moisture loss westward**
Sutlej (सतलुज)–Ganga plain winter rain	**Western Disturbances**

PYQ — UPPCS Prelims 2022, Q22A/R logic: (A) tests same-latitude January contrast. (R) tests lapse-rate wording. Check whether thinner air / height **explains** Agra vs Darjeeling.

Assertion (A): Agra and Darjeeling are located on nearly the same latitude, but the temperature in January in Agra is about 16°C, whereas it is only 4°C in Darjeeling.

Reason (R): Temperature decreases with height due to thinner air compared to places in the plains.

A. (A) false, (R) true

B. (A) true, (R) false

C. Both true, R not the explanation of A

D. Both true and R is the correct explanation of A

Show answer

Ans: D — **(A) true** and **(R) true**; altitude / lapse rate **explains** the January contrast. Same latitude, different height. **C** wrongly denies the causal link. **A/B** falsify correct temperature facts.

Heat budget, albedo, humidity

The Sun sends **short-wave** radiation. The Earth returns **long-wave infrared**. The troposphere is heated mainly from below by that terrestrial long-wave radiation, plus sensible and latent heat. It is **not** heated

mainly by incoming short-wave sunlight.

- **Latent heat:** when water vapour condenses into cloud droplets, heat is released into the air. That heat helps power thunderstorms and monsoon systems.
- Without the natural greenhouse effect (प्राकृतिक ग्रीनहाउस प्रभाव), mean surface temperature would be about **–18 °C**. The actual mean is about **15 °C**.

Surface	Albedo	Fact
Fresh snow	80–90%	**Highest** among usual surface options
Desert sand	30–40%	Moderate
Prairie (प्रेयरी)	20–25%	—
Paddy / cropland	10–20%	Lower
Forest / water (typical)	under 10–20%	Absorb more; warm more

- **Albedo** is the share of sunlight a surface reflects. Fresh snow reflects most. Dark forest absorbs more and warms more.
- **Absolute humidity** is the mass of water vapour in a given volume of air.
- **Relative humidity** is actual vapour as a percentage of what the air can hold at that temperature. When temperature rises, holding capacity rises, so relative humidity falls even if the vapour mass is unchanged.
- Absolute humidity can still rise with stronger evaporation even while relative humidity falls. The two facts are both true, but the second does **not** explain the first.

> **Teaching note:** The troposphere is heated mainly by **long-wave terrestrial radiation (भौमिक विकिरण)** from below. Among common surfaces, **fresh snow** has the highest albedo. Absolute humidity can rise with evaporation while relative humidity falls with warming; the evaporation fact does **not** explain the RH fall.

Same-Latitude / Contrasting Stations Map Bank

Pair / Station	Latitude / Location	What differs	Primary Control
Agra vs Darjeeling	Both ~27°N	January: Agra **16 °C** vs Darjeeling **4 °C**	**Altitude & Lapse Rate** (~1 °C per 165 m; Darjeeling ~2,042 m vs Agra ~169 m)
Amritsar vs Shimla	Both ~31°–32°N	Amritsar hot summer / mild winter vs Shimla snow / cool	**Altitude** (Amritsar ~230 m / 756 ft vs Shimla ~2,400 m / 7,866 ft)
Ajmer vs Shillong	Both ~26°–27°N	Ajmer ~587 mm/yr vs Shillong ~5,160 mm/yr	**Distance from Sea & Orographic Relief** (Khasi Hills funnel vs Thar margin)

Pair / Station	Latitude / Location	What differs	Primary Control
Chennai Station Profile	13°N (Coromandel)	Mean annual temp **26 °C**, annual rain **63 cm**, annual range **9 °C**	**Tropical Maritime & NE Monsoon dependence**

Traditional Indian Seasons (Hindu Calendar Ritus)

According to Indian tradition, the year is divided into 6 bimonthly seasons (*Ritus*):

Sr.	Season (*Ritu*)	Hindu Months	Gregorian Months	Key Characteristics
1	**Vasanta** (वसंत - Spring)	Chaitra – Vaishakha	March – April	Mild pleasant weather, blooming flowers, new foliage
2	**Grishma** (ग्रीष्म - Summer)	Jyeshtha – Ashadha	May – June	Intense heat, dry westerly winds (Loo), localized dust storms
3	**Varsha** (वर्षा - Rainy)	Shravana – Bhadrapada	July – August	Peak South-West monsoon, widespread torrential rains, high humidity
4	**Sharad** (शरद - Autumn)	Ashvina – Kartika	September – October	Retreating monsoon, clear skies, October heat, transition to cool
5	**Hemant** (हेमंत - Pre-winter)	Margashirsha – Pausha	November – December	Pleasant cool weather, Coromandel coast rainfall (NE monsoon)
6	**Shishir** (शिशिर - Winter)	Magha – Phalguna	January – February	Coldest months, frost/fog in north, Western Disturbance rains

> **Chronological Order Drill:** Vasanta (Spring) → Grishma (Summer) → Varsha (Rainy) → Sharad (Autumn) → Hemant (Pre-winter) → Shishir (Winter).

IMD Rainy Day, Water Management & "Thirsty Land" Paradox

- **Official Definition of a Rainy Day (IMD):** A day is officially recorded as a **rainy day** when a meteorological station records ≥ **2.5 mm** of rainfall within a 24-hour period.
- **The "Thirsty Land" Paradox:** Although India receives a generous national average annual rainfall of ~**1,236.4 mm**, it is often described as a "thirsty land". Why? 1. **Temporal concentration:** Nearly 75–90% of rain falls in just 3 to 4 monsoon months (June–September), leaving the country dry for the remaining 8 months. 2. **Rapid runoff:** Torrential downpours cause swift surface runoff into rivers and seas before moisture can infiltrate aquifers. 3. **High evaporation:** High tropical temperatures drive rapid evaporative losses from soil and surface water bodies.
- **Inland Waterways Limitation:** Inland water transport remains underdeveloped largely because rivers outside the snow-fed Himalayas shrink dramatically during the 8 dry months due to monsoon seasonality.
- **Watershed Hallmark in Semi-Arid India:** Embanking seasonal rivulets and rivers into a **system of traditional tanks** and check dams (over 120,000 tanks irrigating >4 million hectares). Tamil Nadu is

the leading state in mandatory rainwater harvesting.

- **Desert Development Programme (DDP):** Launched in 1977–78 to combat desertification, starting with 131 blocks across 21 districts in 5 states (Rajasthan, Gujarat, Haryana, J&K, HP), later covering 235 blocks of 40 districts in 7 states (including Andhra Pradesh and Karnataka).
- **All-India Water Divide:** Major divide separating Arabian Sea drainage from Bay of Bengal drainage, traversing from Western Ghats through Satpura/Vindhya, Aravallis, Delhi ridge, to the Siwaliks and Himalayas.

Temperature inversion

Normally temperature falls with height. In an **inversion**, temperature rises with height near the ground. Cold air below warmer air traps smoke and fog near the surface.

Type	When / where	What happens
Radiation inversion	Clear, calm **winter nights** on the Indo-Gangetic plain	Ground cools fast; fog / smog trap over Delhi, UP, Bihar
Valley / drainage inversion	Kashmir, Doon, hill valleys	Cold air drains to the **floor**; slopes stay milder
Subsidence inversion	Under high-pressure / descending air	Mixing stops; haze can persist

- Inversion suppresses vertical mixing. Pollution stays near the ground. That is why NCR and west-UP winter smog spells persist.

Pressure, wind, Coriolis

Air moves from **high pressure to low pressure**. That pressure difference creates wind.

Earth rotates, so moving air is deflected sideways. That deflection is the **Coriolis effect**. Coriolis does not create wind. It only bends wind that already exists.

Hemisphere	Deflection
Northern (India)	to the **right**
Southern	to the **left**

This is why India's summer monsoon is south-west, not south-east. Southern Hemisphere trade (पण्याध्यक्ष) winds approach the equator from the **south-east**. After they cross into the Northern Hemisphere, Coriolis turns them to the right, so they become **south-west** winds.

Coriolis comes from Earth's **rotation**. It is not magnetism. Heat creates pressure differences; Coriolis only turns the resulting wind.

- In **summer**, hot land creates a thermal **low** over north-west India and Pakistan. Moist ocean air is drawn in and feeds the south-west monsoon.
- In **winter**, cool land creates a thermal **high** over north-west India. Dry winds then blow out from the north-east and north-west.
- The **monsoon trough** is a low-pressure axis along the Himalayan foothills and the Ganga plain.

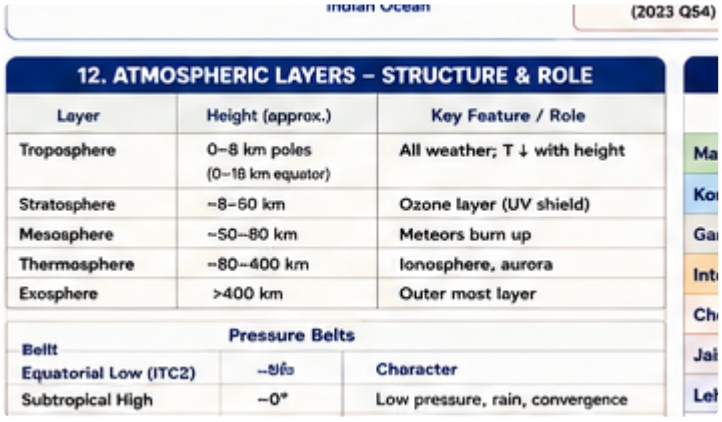
Spell	Trough position	Rain
Active monsoon (सक्रिय)	Over the **Ganga plain**	Plains wet (UP, Bihar, MP)
Break monsoon (विराम मानसून)	Over the **Himalaya**	Central India dry; foothills wet

> **Logic:** leftward deflection in the Southern Hemisphere is **Coriolis**, not temperature, pressure, or the magnetic field.

Atmosphere, pressure belts, ITCZ

The equator receives more sun than the poles. Warm air rises near the equator. Cooler air sinks near about 30° latitude and at the poles. That pattern creates Earth's main **pressure belts**. Air flows between the belts as **planetary winds**, and Coriolis bends those winds.

Layer	Height	Fact
Troposphere (क्षोभमंडल)	0–8 km at poles / 0–18 km at the equator	**All weather**; temperature falls with height
Tropopause	Top of the troposphere	Jets fly near here
Stratosphere	to about 50 km	**Ozone UV shield**
Mesosphere (मध्यमंडल)	about 50–80 km	Meteors burn
Thermosphere	about 80–400 km	Ionosphere and aurora
Exosphere	outermost	Space



Pressure belts and circulation cells

Step	Process	Surface result
1	Strong heating at the equator	Air rises
2	Rising air reduces surface weight	Equatorial **low** / ITCZ
3	Air aloft moves poleward and sinks near 30°	Subtropical **high** (horse latitudes)
4	Surface air returns from 30° to the equator	**Trade winds**

Step	Process	Surface result
5	Other surface air from 30° moves toward 60°	**Westerlies**
6	Cold air sinks at the poles	Polar **high**
7	Polar air moves toward 60° and meets westerlies	Subpolar **low**; **polar easterlies**
Cell	Span	Role
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Hadley cell	Equator to ~30°	Rise at ITCZ, sink at 30°, trades return
Ferrel cell	~30° to 60°	Mid-latitude belt of the westerlies
Polar cell	~60° to pole	Polar easterlies; rise near 60°
Nickname	Meaning	
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Doldrums	Rainy, weak-surface-wind belt of the ITCZ	
Horse latitudes	Dry, calm subtropical highs near 30°	
Belt	Latitude	Pressure
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Equatorial low / **ITCZ**	Near 0° (shifts with the Sun)	**Low**
Subtropical high	About **30° N / S**	**High**
Subpolar low	About **60° N / S**	**Low**
Polar high	About **90°**	**High**

ITCZ

The **Inter Tropical Convergence Zone** is the rainy low-pressure belt where Northern Hemisphere and Southern Hemisphere trade winds meet.

Trade winds move toward the heated equatorial belt, converge, and rise. Rising moist air cools and gives heavy rain. The ITCZ is therefore a **low**, never a high.

It is not fixed on the equator. It shifts with the overhead Sun.

Season	ITCZ position	Effect on India
Summer (Jun–Sep)	Shifts **north** over South Asia	Forms / guides the monsoon trough; supports SW monsoon
Winter (Dec–Feb)	Shifts **south**	Most of India under dry NE trades

Season	ITCZ position	Effect on India
Strong El Niño years	Indian branch often weaker	Monsoon tends to weaken

If the monsoon trough lies on the Ganga plain, rains are **active**. If it shifts onto the Himalaya, a **break** occurs and central India dries.

Planetary winds

Wind	Direction	Where	Cause	Effect
Trade winds	East → west (NE trades in NH; SE trades in SH)	Between ~30° highs and the ITCZ	Air leaves subtropical high toward equatorial low; Coriolis turns it	Feed ITCZ rain; after equator crossing they become India's SW monsoon
Westerlies	West → east	About **30°–60°** (both hemispheres)	Air leaves 30° high toward 60° low; Coriolis turns it	Drive mid-latitude storms and help steer Western Disturbances
Polar easterlies	East → west	Polar high toward ~60°	Cold air leaves polar high; Coriolis turns it	Bring cold outbreaks; meet westerlies at the subpolar low

Winter NE trades over India are mostly dry on land. Summer SE trades that cross the equator and turn right become the **south-west monsoon**. Westerlies exist in **both** hemispheres; they are not Southern Hemisphere only.

> **Logic:** Protective ozone sits in the **stratosphere**, not the troposphere. The ITCZ is a **low**, not a high. Ozone blocks UV **and** depletion is linked to CFCs — both statements hold.

2.2 Monsoon System

A monsoon is a seasonal reversal of wind. In summer, heated land draws moist ocean air inland. In winter, cooled land sends dry air outward.

- The word **monsoon** comes from Arabic **mausim** (season).
- India receives about **75–90%** of its rain from the south-west monsoon (June–September).
- India's climate has both tropical and subtropical traits. The Himalaya adds climatic distinctiveness. India is not limited to the Tropics; it extends to about **37°6′ N**.
- The monsoon is produced by land–sea heating, the northward shift of the ITCZ, cross-equatorial flow, the Mascarene High, the Somali Jet, and Himalayan / Tibetan heating. The Himalaya helps, but it is not the only cause.

System	Months	Where the rain falls	Trap
South-west monsoon	Jun–Sep	Most of India (~75–90% of annual rain)	TN SE coast stays **dry**
Retreating monsoon	Sep–Nov	Withdrawal phase with **October heat** and BoB / Andaman Sea cyclones	Not WD winter rain
North-east monsoon	Oct–Dec	**TN, south AP, SE Karnataka fringe, Puducherry, Karaikal**	Not all-India winter rain
Western Disturbances	Nov–Feb (peak)	NW India rain/snow	Extra-tropical; **not** a monsoon

Monsoon mechanism

Two standard explanations work together:

Concept	Fact
Thermal	Land heats faster than sea → summer low over India pulls ocean air in
Dynamic (Flohn)	Pressure belts and the ITCZ shift with the Sun; monsoon is largely that seasonal shift
Step	What happens
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1	Tibetan Plateau / NW India heat strongly in late spring
2	ITCZ shifts **north** onto India
3	**Mascarene High** strengthens over the southern Indian Ocean
4	SE trades cross the equator roughly between **40° E and 60° E**; Coriolis turns them SW
5	Cross-equatorial flow strengthens as the **Somali Jet**
6	STWJ jumps north of Tibet; **TEJ** sets up near 14° N
7	Onset **burst** at Kerala (~1 June), then the front marches north

- In winter the ITCZ shifts south. Winds reverse to the north-east. Over most of India those winds are dry. They rain on the Coromandel only after picking up Bay moisture.
- Withdrawal begins in the **north-west** and ends in the extreme south. That north-to-south pullback is the retreating monsoon.

Concepts of Monsoon Origin & Upper Atmosphere Controls

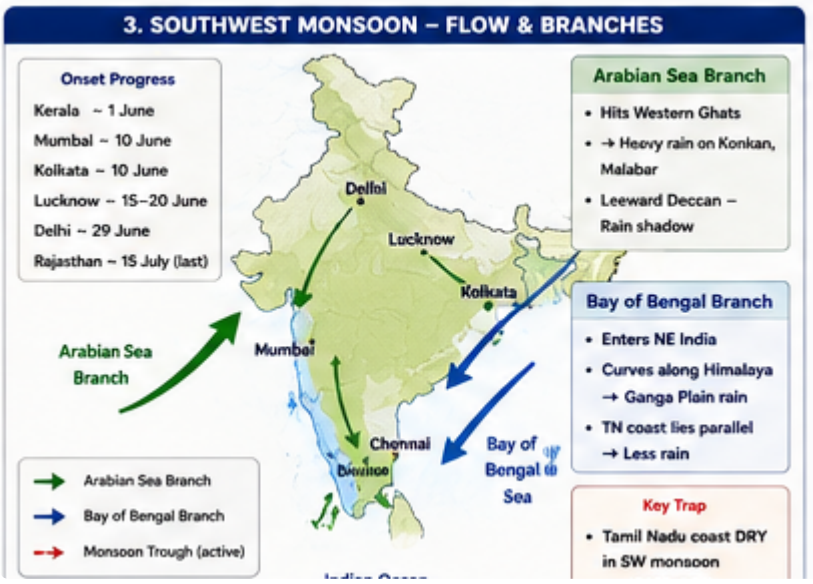
1. **Thermal Concept (Halley / Classical):** Differential heating rates between the vast continental landmass of Asia/India and the surrounding Indian Ocean drive an enormous seasonal sea-breeze / land-breeze cycle.
2. **Dynamic Concept (Flohn):** The monsoon is primarily the seasonal displacement and migration of the planetary wind belts and the Inter-Tropical Convergence Zone (ITCZ) northward across the equator in summer and southward in winter.

3. **Jet Stream Theory & Tibetan Plateau:** - The high **Tibetan Plateau** acts as an elevated mechanical barrier and summer thermal engine, dividing the Subtropical Westerly Jet (STWJ) into two branches:
    - **Northern branch:** Flows north of the Himalayas/Tibetan Plateau across Central Asia and China; it plays **no role** in India's winter precipitation.
    - **Southern branch:** Flows south of the Himalayas across northern India; it steers Mediterranean **Western Disturbances** and governs winter weather.
    - In late May/early June, the sudden northward shift of this southern branch north of Tibet triggers the dramatic **"burst"** of the South-West monsoon over the Indian peninsula.

Drainage & Regional Rainfall Traps

- **Chhattisgarh Basin & Arabian Sea Branch:** The Arabian Sea branch splits after hitting the Western Ghats (one sub-branch heads along the Aravallis, another up the Narmada-Tapi into MP and western CG). The inland **Chhattisgarh Basin** remains largely sheltered from this branch; it receives almost its entire rainfall from the **Bay of Bengal branch**. Its climate is classified as **Humid South-East**.
- **Pushkar Hills Heavy Rain → Balotra Floods:** When intense rainfall strikes the Pushkar hills in Ajmer, floodwaters channel down the Luni river basin and surge into **Balotra (Barmer)** because Balotra sits on the low downward gradient of the Pachpadra depression.
- **Signs of Monsoon Decline (Retreating Monsoon):** 1. **Clear sky** replaces monsoon cloudiness. 2. **October Heat:** Rising daytime temperatures on land combined with high residual humidity. 3. **Low pressure shift to Bay of Bengal:** Development of cyclonic depressions and storms that track across the Coromandel coast.
- **Rainfall Gradient across Major Cities (Descending Order):**
- **Kochi (~3,932 mm)** > **Kolkata (~2,451 mm)** > **Patna (~1,100 mm)** > **Delhi (~904 mm)**.
- **Lowest Rainfall Benchmarks:**
- **Leh (Ladakh UT):** ~4.17 cm (41.7 mm) — **lowest in all India**.
- **Jaisalmer:** ~31.67 cm — lowest in Rajasthan.
- **Bikaner:** ~32.84 cm.

Southwest monsoon



- The south-west monsoon lasts from **June to September**. It gives about **75–90%** of India's annual rain.
- IMD's normal onset at **Kerala** is about **1 June** (first among major states). The front then moves north at roughly ten days per degree of latitude.

Place	Normal onset	Withdrawal (start)
Kerala	about 1 June	Extreme south last (December for NE rains)
Mumbai / Kolkata	about 10 June	Early October
Lucknow (लखनऊ) / eastern UP	about 15–20 June	Late September
Delhi	about **29 June**	Late September
Rajasthan	about **15 July** (last)	September
Chennai	South-west is **weak and dry**	North-east monsoon October–December

Two branches — remember as two arms of the same monsoon

Branch	Path	Wet places	Dry / weak places
Arabian Sea branch	Arabian Sea → Western Ghats → inland (also a stream toward central India / Ganga that later meets the Bay branch)	Konkan, Malabar, Goa (गोवा); some rain into Gujarat / western MP	Leeward Deccan and TN rain shadow; **Chhattisgarh basin** is little fed by this branch
Bay of Bengal branch	BoB → NE India → curves west along Himalaya	Assam, Meghalaya, Ganga plain, UP, Bihar	TN coast (sits **parallel** — little orographic lift)

- The Tamil Nadu coast lies **parallel** to the Bay branch. There is little orographic lift. That is why the Tamil Nadu coast stays **dry in the south-west monsoon**.
- The **burst** is the sudden heavy rain after May heat. Temperature then falls.
- Bay of Bengal monsoon depressions usually track west-north-west into Madhya Pradesh, Uttar Pradesh, and Rajasthan.

Rainfall character facts (Ganga / all-India)

Pattern	Fact
Season	Mostly **June–September** (SW monsoon)
Relief	Orography governs amounts (Western Ghats, Meghalaya scarp)
Distance from sea	Rain generally **declines inland** / with distance from the moisture source
Northern plains	Annual rain **decreases east** → **west** (humid Bay air loses moisture westward)
Duration	Monsoon season length **decreases south** → **north** (Kerala longer than the far north-west)
Break	Rainless spells when the trough sits on the **Himalaya**

Pattern	Fact
Rajasthan dryness	Aravalli runs **roughly parallel** to SW winds → little forced uplift on the western Thar

PYQ — UPPCS Prelims 2023, Q54A/R logic: (A) tests TN dryness in SW season. (R) packs two causes — parallel to Bay branch **and** Arabian Sea rain shadow. Check if both halves of R are true and whether they explain A.

Assertion (A): The Tamil Nadu coast remains dry during the South-West monsoon season.

Reason (R): The Tamil Nadu coast is situated parallel to the Bay of Bengal branch of the South-West monsoon and it lies in the rain shadow area of the Arabian Sea branch of the South-West monsoon during the monsoon season.

A. (A) true, (R) false

B. (A) false, (R) true

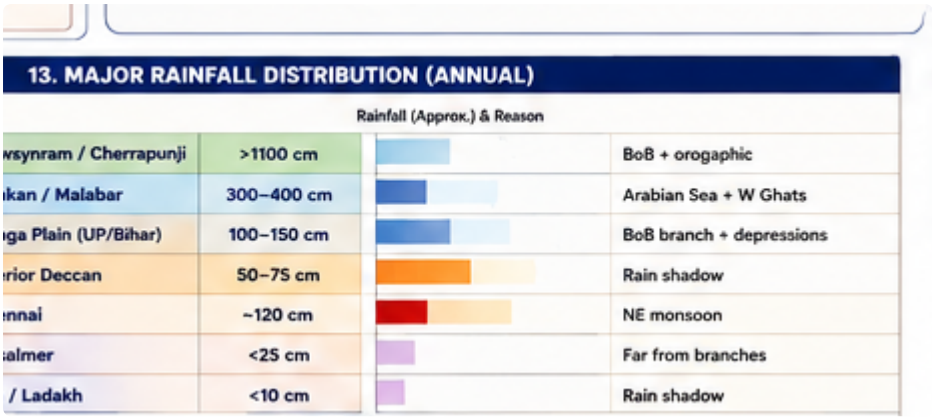
C. Both true, R not the explanation

D. Both true and R is the correct explanation

Show answer

Ans: D — **(A) true** and **(R) true**; orientation + rain shadow **explain** TN SW-monsoon dryness. **C** denies the causal link. **A/B** falsify correct monsoon geography.

Zone	Rain (approx.)	Why
Mawsynram / Cherrapunji (**Khasi (खासी) Hills**, Meghalaya)	more than 1100 cm	Bay branch plus orography; Mawsynram = wettest class
Konkan / Goa	300–400 cm	Arabian Sea plus Western Ghats
Ganga plain (UP / Bihar)	100–150 cm	Bay branch plus depressions
Interior Deccan	50–75 cm	Rain shadow
Chennai	about 120 cm, **mostly north-east monsoon**	South-west is dry
Jaisalmer	less than 25 cm	Far from both branches; Aravalli parallel
Leh / Ladakh	less than 10 cm (often ~4 cm class)	Himalayan rain shadow — driest among usual options



13. MAJOR RAINFALL DISTRIBUTION (ANNUAL)

		Rainfall (Approx.) & Reason
wsynram / Cherrapunji	>1100 cm	BoB + orogaphic
kan / Malabar	300–400 cm	Arabian Sea + W Ghats
ga Plain (UP/Bihar)	100–150 cm	BoB branch + depressions
rior Deccan	50–75 cm	Rain shadow
nnai	~120 cm	NE monsoon
almer	<25 cm	Far from branches
/ Ladakh	<10 cm	Rain shadow

- All-India mean rainfall is about **118–124 cm** (class figures vary by yearbook; ~**118 cm / ~1236 mm** both appear).
- IMD calls a **rainy day** a day with **2.5 mm or more** rain in 24 hours.
- India's tropical monsoon belt allows **kharif (खरीफ), rabi (रबी), and zaid (जायद)**. The claim that India has the highest cultivated area versus the USA, China, and Japan is false.

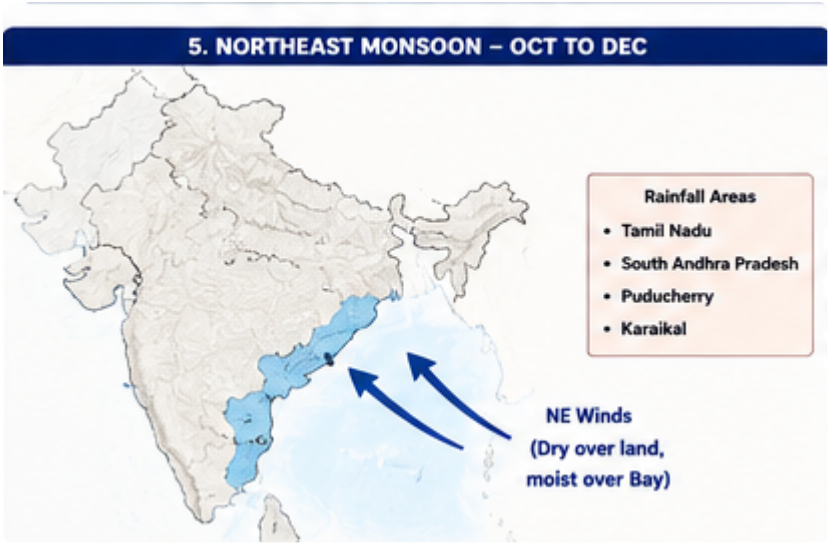
How rain forms (three types)

Type	Cause	India example
Orographic	Moist wind is forced **up a mountain**, cools, and rains on the windward side	Western Ghats windward vs Deccan; Meghalaya scarp
Convectional	Hot ground heats air → air rises in a column → afternoon thunder showers	Pre-monsoon storms; parts of interior summer heating
Cyclonic / frontal	Air rises in a **low-pressure storm** system	Monsoon depressions; Western Disturbances; Bay cyclones

- On the orographic path, the **leeward** side stays dry (rain shadow).
- Cyclonic / frontal lift is the storm-system path for monsoon depressions, Western Disturbances and Bay cyclones.

Map line	Meaning
Isotherm	Same temperature
Isobar	Same pressure
Isohyet	Same rainfall

Northeast monsoon and retreating monsoon



These two names cover the same autumn window, but they are not the same process.

Feature	Retreating SW monsoon	North-east monsoon
Months	About **Sep–Nov**	About **Oct–Dec**
Process	SW monsoon withdraws from north to south	Dry NE winds pick up Bay moisture and rain on the SE coast
Signature	October heat, clear spells, and Bay / Andaman Sea cyclones	Coromandel rain
Rain belt	Withdrawal rains + east-coast cyclone rains	TN, south AP, SE Karnataka, SE Kerala fringe, Puducherry, Karaikal
North India	Mostly dry	Mostly dry

- Signs of retreat include clearer skies, rising land temperatures, and cyclone activity shifting into the Bay.
- The Coromandel coast gets about half to two-thirds (ते-भागा) of its annual rain in this season.
- Mumbai and Gujarat lose SW rains and do not receive the NE monsoon, so October–November is dry there.

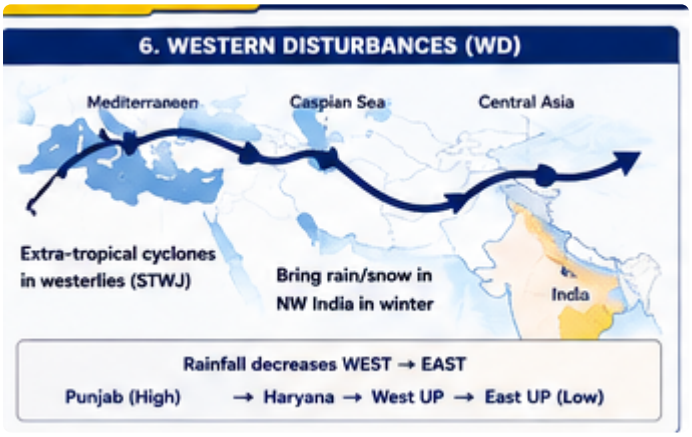
Indian seasons (IMD)

Season	Months	Main weather
Cold weather	December–February	Western Disturbance rain/snow in the north-west, fog on the plains, and the rabi season
Hot weather	March–May	Loo heat waves, Nor'westers, and mango / blossom showers
South-west monsoon	June–September	Main rainy season for most of India

Season	Months	Main weather
Retreating	October–November	Withdrawal, October heat, Bay cyclones, then NE monsoon on the SE coast

- **Cold weather (Dec–Feb):** north-west rain and snow come from **Western Disturbances**. Fog is common on the Indo-Gangetic plain. This is the main **rabi** (रबी) wheat season. Most of India stays dry under **NE trade / land winds**.
- **Hot weather (Mar–May): Loo** heat waves strike the north-west plains. **Nor'westers** and other pre-monsoon storms hit the east. **Mango** and **blossom** showers appear in the south.
- **South-west monsoon (Jun–Sep):** This is the main rainy season. Tamil Nadu's **south-east coast** stays relatively dry in this phase.
- **Retreating monsoon (Oct–Nov)** is the withdrawal of the southwest monsoon from the north-west.
- The same season brings **October heat** and Bay of Bengal cyclones.
- The **north-east monsoon** then begins to rain on the Tamil Nadu coast.

Western Disturbances



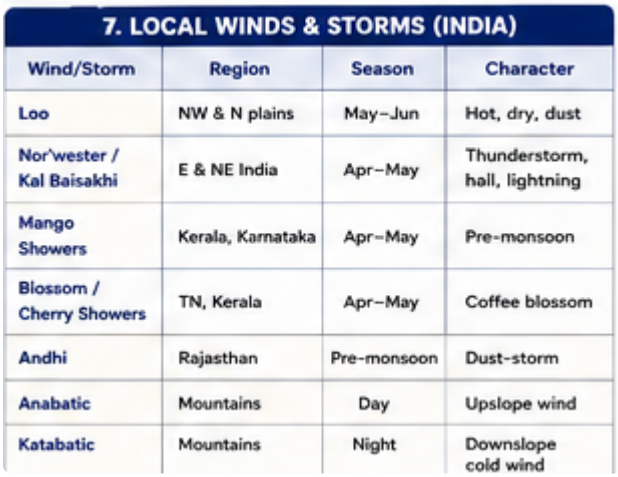
Western Disturbances are **extra-tropical** winter storms. They form near the Mediterranean–Caspian region, travel east in the subtropical westerly jet, and bring rain or snow to north-west India. They are not monsoon lows and not Bay of Bengal depressions.

Item	Note
Type	Extra-tropical cyclone
Carrier jet	STWJ branch **south of the Himalaya** (not the northern Tibetan branch)
Peak season	November–February
Rain / snow belt	Punjab, Haryana, Himachal, J&K / Ladakh fringe, Uttarakhand, western UP, northern Rajasthan
Gradient	Decreases **west** → **east**
Farm link	Helps rabi wheat; also hail, fog, avalanche risk

> **Logic:** ** Winter north-west rain comes from **Western Disturbances**. The rainfall gradient runs **west to east**.

2.3 Winds, Storms and Cyclones

Local winds and storms of India



7. LOCAL WINDS & STORMS (INDIA)			
Wind/Storm	Region	Season	Character
Loo	NW & N plains	May–Jun	Hot, dry, dust
Nor'wester / Kal Baisakhi	E & NE India	Apr–May	Thunderstorm, hail, lightning
Mango Showers	Kerala, Karnataka	Apr–May	Pre-monsoon
Blossom / Cherry Showers	TN, Kerala	Apr–May	Coffee blossom
Andhi	Rajasthan	Pre-monsoon	Dust-storm
Anabatic	Mountains	Day	Upslope wind
Katabatic	Mountains	Night	Downslope cold wind

Local winds are short-lived and stay near the ground.

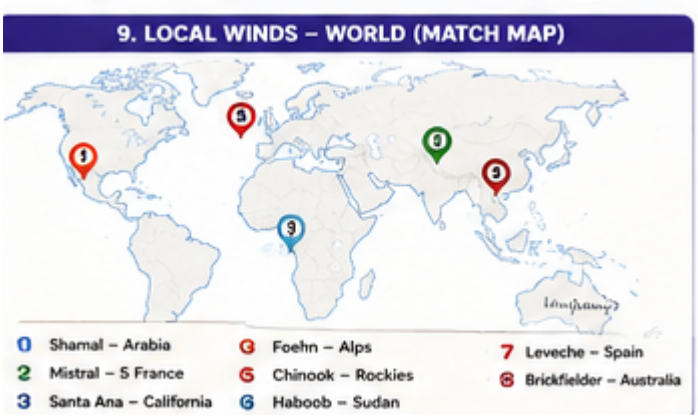
- The **Loo** is a hot, dry, dusty wind on the plains of Rajasthan, Punjab, Haryana, Uttar Pradesh, and Bihar in **May–June**. It is a heat-wave wind, not a thunderstorm.
- The **Nor'wester / Kal Baisakhi** is a violent pre-monsoon thunderstorm over West Bengal, Assam, Odisha, Bihar, and Jharkhand in **April–May**.
- **Mango showers** are April–May pre-monsoon rains in Kerala and Karnataka that help mangoes ripen.
- **Blossom / cherry showers** are April–May pre-monsoon rains in the coffee belt of Tamil Nadu and Kerala.
- **Andhi** is a pre-monsoon dust-storm wall of Rajasthan.
- **Elephanta** (एलिफेंटा) is a Malabar wind of about September at the fading of the SW monsoon.

Time	Land vs sea	Wind
Day	Land heats faster → low over land	**Sea breeze** (sea → land)
Night	Land cools faster → high over land	**Land breeze** (land → sea)
Wind	When	Direction
------	------	----------
Anabatic	Day	Upslope
Katabatic	Night	Downslope
Foehn / Chinook	After crossing a mountain	Warm and dry downslope

Foehn / Chinook are warm-dry lee winds. They are not the same as cold night-time katabatic drainage.

Name	Region	Season	Character
Loo	RJ, PB, HR, UP, Bihar plains	May–June	Hot, dry dust and heat-wave; **not** a thunderstorm
Nor'wester / Kal Baisakhi	WB, Assam, Odisha, Bihar, Jharkhand	April–May	Violent **thunderstorm**; hail, lightning
Mango showers	**Kerala, Karnataka**	April–May	Pre-monsoon; mango ripening
Blossom / cherry showers	TN, Kerala (coffee belt)	April–May	Coffee flowering
Andhi	Rajasthan	Pre-monsoon	Dust-storm wall
Elephanta	Malabar / Kerala	About September	End of SW monsoon; rough sea
Sea / land breeze	Coasts	Daily	Onshore by day; offshore by night
Anabatic	Mountain slopes	**Day**	**Upslope**
Katabatic	Valleys	**Night**	**Downslope** cold drainage

World local winds (full match map)



- Past papers have already asked Mistral, Shamal, Chinook, Foehn, Santa Ana, Haboob, Brickfielder, Leveche, Black roller, and Yamo. Other standard local winds still worth knowing include **Sirocco, Bora (बोरा), Harmattan (हरमट्टन), Khamsin (खामसिन), Zonda, Samun**, and the Willy-willy trap.
- Local winds are short-lived and near the ground. Group them by **family**, then fact the region.
- **Warm and dry after crossing a mountain (Foehn family)**
- The **Foehn (Föhn)** is the Alpine warm, dry downslope. It can melt snow and help grapes ripen.
- The **Chinook** is the Rockies twin. It is warm **and dry**. Ranchers call it the **snow eater**.
- The **Zonda** is the same family on the eastern Andes in **Argentina**.
- The **Berg** wind is the South African cousin.
- The **Santa Ana** is the hot, dry California downslope.
- New Zealand's **Nor'wester** (Canterbury) is a foehn-type wind. It is **not** the Indian Kal Baisakhi thunderstorm.

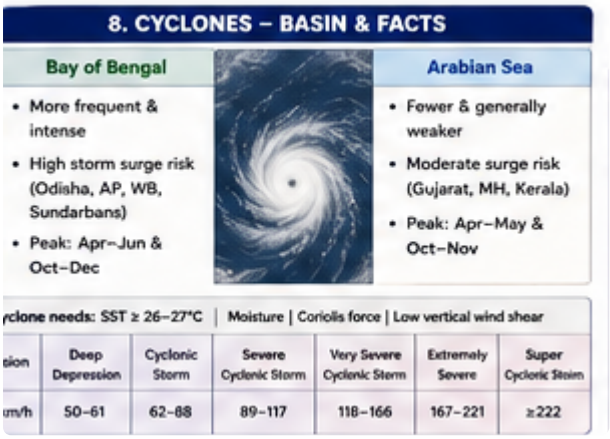
- **Cold downslope / outbreak winds**
- The **Mistral** is a cold, dry blast down the **Rhône** in southern **France**, funnelled between the Alps and the Massif Central. It is **not** Australian.
- The **Bora** is the cold north-easterly cousin on the **Adriatic** (Italy–Croatia).
- The **Tramontane** is a cold north wind of southern France and the western Mediterranean.
- The **Gregale** is a cold north-east wind around **Malta**.
- The **Pampero** is a cold outbreak on the Argentine pampas.
- The **Buran / Purga** is a Siberian blizzard-type wind.
- The **Norte** is a cold outbreak into Mexico and the Gulf.
- **Hot, dusty desert winds**
- The **Sirocco** is a hot, dusty Saharan wind that crosses into the **Mediterranean** and southern Europe. It can raise “blood rain” from red dust.
- The **Khamsin** is Egypt (मिस्र)’s sirocco cousin (about fifty days of spring heat).
- The **Ghibli** is the Libyan name. The **Chili** is the Tunisian name. The **Leveche** is the Spanish name (2021 correct pair).
- The **Shamal** is a dusty north-westerly of **Arabia and the Persian (फ़ारसी) Gulf**. It is **not** Austrian.
- The **Simoom** is a scorching Arabian desert wind.
- The **Harmattan** is a dry, dusty winter wind from the Sahara over **West Africa** and the Gulf of Guinea. It is **not** an East African coast wind.
- The **Samun** is a hot wind of **Kurdistan / Iran** (UPSC (संघ लोक सेवा आयोग) 2001-type match).
- The **Brickfielder** is a hot wind of **Australia**.
- The **Black roller** is a dusty North American plains wind.
- The **Haboob** is a dust-wall storm of **Sudan** and the wider Sahara.
- The **Karaburan** (“black storm”) is a dusty Central Asian wind.
- The **Loo** is the north-Indian plains member of this hot family. It lives in the India table above.
- **Other named winds used in match lists**
- The **Levanter** is an easterly through the Strait of **Gibraltar**.
- The **Etesian / Meltemi** is a summer northerly of the **Aegean**.
- The **Yamo** is a Japanese local wind (2019 correct pair).
- The **Cape Doctor** is a south-easter at Cape Town.
- The **Williwaw** is a violent squall of Alaska / Magellan.
- **Do not mix cyclone names into this list**
- **Willy-willy** is an Australian **tropical cyclone**. It is **not** a local wind like the Brickfielder.
- Hurricane, typhoon, taifu, and baguio are **synoptic storms**. They belong in the cyclone-naming table below.

Family	Wind	Region	trap
Warm dry downslope	**Foehn**	Alps	Not Australia
Warm dry downslope	**Chinook**	Rockies	Not a cold wind
Warm dry downslope	**Zonda**	Andes, Argentina	Not Alps

Family	Wind	Region	trap
Warm dry downslope	**Santa Ana**	California	Not Spain
Warm dry downslope	**Berg**	South Africa	—
Cold	**Mistral**	S France, Rhône	**Not Australia**
Cold	**Bora**	Adriatic	Not Chinook
Cold	**Pampero**	Argentina	—
Cold	**Buran / Purga**	Siberia	—
Hot dusty	**Sirocco**	Sahara → Mediterranean	—
Hot dusty	**Khamsin**	Egypt	Not Australia
Hot dusty	**Shamal**	Arabia / Gulf	**Not Austria**
Hot dusty	**Harmattan**	**West** Africa	Not East Africa
Hot dusty	**Samun**	Kurdistan	—
Hot dusty	**Leveche**	Spain	—
Hot dusty	**Ghibli**	Libya	—
Hot dusty	**Brickfielder**	Australia	Not Mistral
Hot dusty	**Haboob**	Sudan	—
Hot dusty	**Black roller**	North America	—
Other	**Yamo**	Japan	—
Other	**Levanter**	Gibraltar	—
Not a local wind	**Willy-willy**	Australia	Cyclone, not Brickfielder

> **Logic:** Chinook is warm and dry **and** Foehn is in the Alps. Mistral is France. Shamal is Arabia. Harmattan is **West** Africa. Willy-willy is a **cyclone**.

Tropical cyclones



A tropical cyclone is a large spinning storm over warm ocean. It needs sea-surface temperature of about **26–27 °C or more**, deep moisture, Coriolis spin, and low vertical wind shear. Coriolis is too weak on the equator, so genesis is usually between about **5° and 30°**.

Need	Note
Sea surface temperature	About **26–27 °C or more**
Moisture	Deep moist air
Coriolis	Enough spin — usually **5°–30°** latitude
Vertical wind shear	**Low** (strong SW-monsoon shear suppresses many Bay storms in Jul–Sep)
Part	Meaning
------	---------
Eye	Calm centre with sinking air and clearer skies
Eyewall	Ring of strongest winds and rain around the eye
Spiral bands	Outer rain bands

- Peak seasons are pre-monsoon (May–June) and post-monsoon (October–December).
- In the Northern Hemisphere these storms spin anti-clockwise.

Basin	Frequency / strength	Why	Hit states
Bay of Bengal	**More and stronger**	Warm shallow shelf + river freshwater; storm surge (तूफानी महोर्मि) risk high	Odisha, AP, West Bengal, Sundarbans (सुंदरबन)
Arabian Sea	Fewer, often weaker	Higher salinity; narrower fetch	Gujarat, Maharashtra, Kerala still take hits
IMD class	Wind (km/h)		
----------	----------		

Basin	Frequency / strength	Why	Hit states
Depression	31–49		
Deep depression	50–61		
Cyclonic storm	62–88		
Severe cyclonic storm	89–117		
Very severe	118–166		
Extremely severe	167–221		
Super cyclone	**222 or more**		
Local storm	Scale	Fact	
---	---	---	
Thunderstorm / Nor'wester	Hours; local convection	**Not** a cyclone	
Tropical cyclone	Days; ocean-scale	Synoptic storm	

Cyclone naming (match bank — next year can swap any pair)

- The **WMO/ESCAP** panel agrees the name lists. **RSMC New Delhi (IMD)** assigns names in the north Indian Ocean.
- Devastating names are **retired**. India has contributed names such as **Amphan**, **Yaas**, **Fani**, **Vayu**, and **Nisarga**.
- UPPCS has already asked the four (चातुर्याम) classic pairs (Hurricane / Typhoon–Taifu / Baguio / Willy-willies). Next stem can still pick **Cyclone** (NIO), **Cordonazo**, or split Taifu from Typhoon. Learn the **full basin bank**.

Local / basin name	Where	Note
Hurricane ★	**USA** / Atlantic–Caribbean / E Pacific	2019–2020 PYQ fact
Typhoon ★	NW Pacific — **China** / Philippines / Taiwan class	2020 Q74 "Typhoons–China"
Taifu ★	**Japan**-facing name for NW Pacific storm	Same basin as typhoon; 2019 paired with Japan
Baguio / Baguios ★	**Philippines**	2019–2020 fact
Willy-willies ★	**Australia**	Cyclone — **not** Brickfielder (hot local wind)
Cyclone	North Indian Ocean — **India / BoB / Arabian Sea**	IMD / RSMC New Delhi names

Local / basin name	Where	Note
Cordonazo	West Mexico / E Pacific coast	Lucent-class distractor
Hurricane (also)	Caribbean / Gulf of Mexico	Same Atlantic family as USA

Named-event note (not a closed four-pack): Amphan (2020, BoB→WB/Odisha), Fani (2019), Yaas (2021), Tauktae (2021, Arabian), Biparjoy (2023), Hudhud (2014), Phailin (2013), Odisha Super Cyclone (1999).

> **Logic:** Willy-willies–Australia, Taifu–Japan, Baguio–Philippines, Hurricanes–USA. Alternate set: Baguios–Philippines, Hurricanes–USA, Typhoons–China, Willy-willies–Australia. Willy-willy ≠ Brickfielder.

2.4 Classification and ocean–atmosphere phenomena

Köppen (India) versus Thornthwaite



Köppen classifies climate by temperature and rainfall letter codes. Thornthwaite classifies by moisture balance. The line **"Vegetation is the true index of climate"** belongs to Thornthwaite, not Köppen.

Thinker	Method	Note
Köppen	Temperature + precipitation **letter codes**	India's Am / Aw / As / BWh / Cwg map
Thornthwaite	Moisture balance (P minus potential ET)	**"Vegetation is the true index of climate"** — 2018 Q25
Trewartha	Modified Köppen	**Not** Thornthwaite

How to read Köppen letters (India map)

Letter clue	Meaning	India example
A	Tropical	Kerala, Deccan, Coromandel
B	Dry (arid / semi-arid)	Thar, rain-shadow Deccan fringe
C	Temperate / subtropical with distinct seasons	Ganga plain
m	Monsoon (short dry spell)	Kerala / Konkan / NE hills
w	Dry **winter**	Interior Deccan (**Aw**); Ganga plain (**Cwg**); also the second small letter in strings like **Amw**
s	Dry **summer**	**Tamil Nadu As** — dry in SW monsoon; rain in NE monsoon
h	Hot	Thar **BWh**

- In fuller Köppen strings, the **second small letter** marks the **seasonality of dryness**. **Amw** still maps to the **humid monsoon west coast and parts of the north-east** on India maps — not Thar (**BWh**) and not Coromandel dry-summer (**As**).

Code	Type	India
Am / Amw	Tropical monsoon (short dry spell; **w** = dry winter in full strings)	Kerala / Konkan, north-east (Cherrapunji)
Aw	Tropical savanna (सवाना)	Interior Deccan
As	Monsoon with dry **summer**	**Tamil Nadu / Coromandel** (rain in NE monsoon)
BWh	Hot desert	**Thar** (थार), Kutch
BSh	Semi-arid	Punjab, Haryana, Gujarat, rain-shadow Deccan
Cwg	Monsoon with dry winter	**Indo-Gangetic / Great Plains** (Lucknow, Delhi, Patna (पटना)); also covers **North Bihar**, **North Bengal / much of NE plain** on class maps
H	Highland	Himalayan slopes

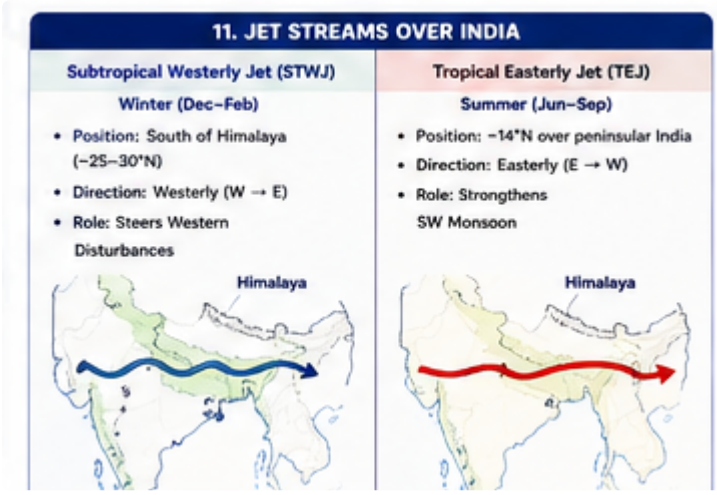
Code	Type	India
E	Polar / tundra (टुंड्रा)	Highest snow zones only

- India is **not** all **Am**. The interior Deccan is **Aw**. The Thar is **BWh**. The Tamil Nadu coast is often mapped as **As** because its dry season is the south-west-monsoon **summer**.
- The "humid south-east" belts (Chotanagpur (छोटानागपुर)–Odisha–south Chhattisgarh fringe) sit in the wetter monsoon east; do not confuse with Thar **BWh**.

Köppen Code	Climatic Type	Representative Regions in India
Amw	Monsoon with short dry season	Western coastal plains south of Goa (Malabar/Konkan), parts of Arunachal/Assam
As	Monsoon with dry summer	**Coromandel Coast** (coastal Tamil Nadu and adjoining Andhra Pradesh)
Aw	Tropical Savanna (wet & dry)	Most of the Peninsular plateau south of the Tropic of Cancer
BWhw	Tropical desert (arid)	Western Rajasthan (Thar Desert)
BShw	Semi-arid steppe	Rain-shadow belt east of Western Ghats; parts of Rajasthan, Haryana, Punjab
Cwg	**Humid subtropical with dry winter**	**Great Plains of India (Indo-Gangetic plain), North-East India (including North Bengal and Assam plains), and North Bihar**
Dfc	Cold humid winter with short summer	High-altitude Sikkim and Arunachal Pradesh Himalayas
E / ET	Polar / Tundra	High mountain belts of Jammu & Kashmir, Ladakh, Himachal Pradesh, Uttarakhand

> **Key UPPCS / UKPCS Trap: Cwg** is the exact Köppen code for the Great Plains of India, North Bihar, and North-East India (including North Bengal).

Jet streams



Jet streams are narrow, fast winds near the tropopause. Most mid-latitude jets are **westerly** (west → east). Speeds of about **300–500 km/h** are typical. The summer **Tropical Easterly Jet** over India is the main exception.

Jet	Season	Direction	Where	Role for India
Polar front jet	Year-round (shifts)	Westerly	High latitudes	Extra-tropical weather belt
STWJ	Winter	Westerly	About 25–30° N, south of Himalaya	Steers Western Disturbances
STWJ north branch	Winter	Westerly	North of Tibet after the plateau splits the jet	Does not drive India's winter rain
STWJ summer position	Summer	Westerly	North of Tibet	Clears space for monsoon setup
TEJ	Summer	Easterly	Near 14° N over the peninsula	Supports SW monsoon circulation
Somali Jet	Summer	Cross-equatorial / SW	Off East Africa → Arabian Sea	Feeds SW monsoon moisture

Calling jet streams generally easterly is false. In winter the Asian westerly jet is split by Tibet; India's Western Disturbances ride the **southern** branch.

- The **Tropical Easterly Jet (TEJ)** is a **summer** upper-level **easterly** near about **14° N** over the peninsula.
- It helps maintain the **southwest-monsoon circulation** after the STWJ has jumped north of Tibet.
- It is **not** the winter jet that steers Western Disturbances, and it is **not** proof that "all jets are easterly."
- Keep TEJ separate from the **Somali Jet**, which is the low-level cross-equatorial moisture feed into the Arabian Sea.

PYQ — UPPCS Prelims 2024, Q60A/R logic: (A) claims jets are high-altitude **easterlies** (false as a general rule). (R) states a plausible speed. Even if R is true, it does **not** make A true.

Assertion (A): Jet streams discovered during World War II are high altitude easterly winds.

Reason (R): Jet streams flow with a speed of 300–500 km/hour.

A. Both true and R explains A

B. (A) false, (R) true

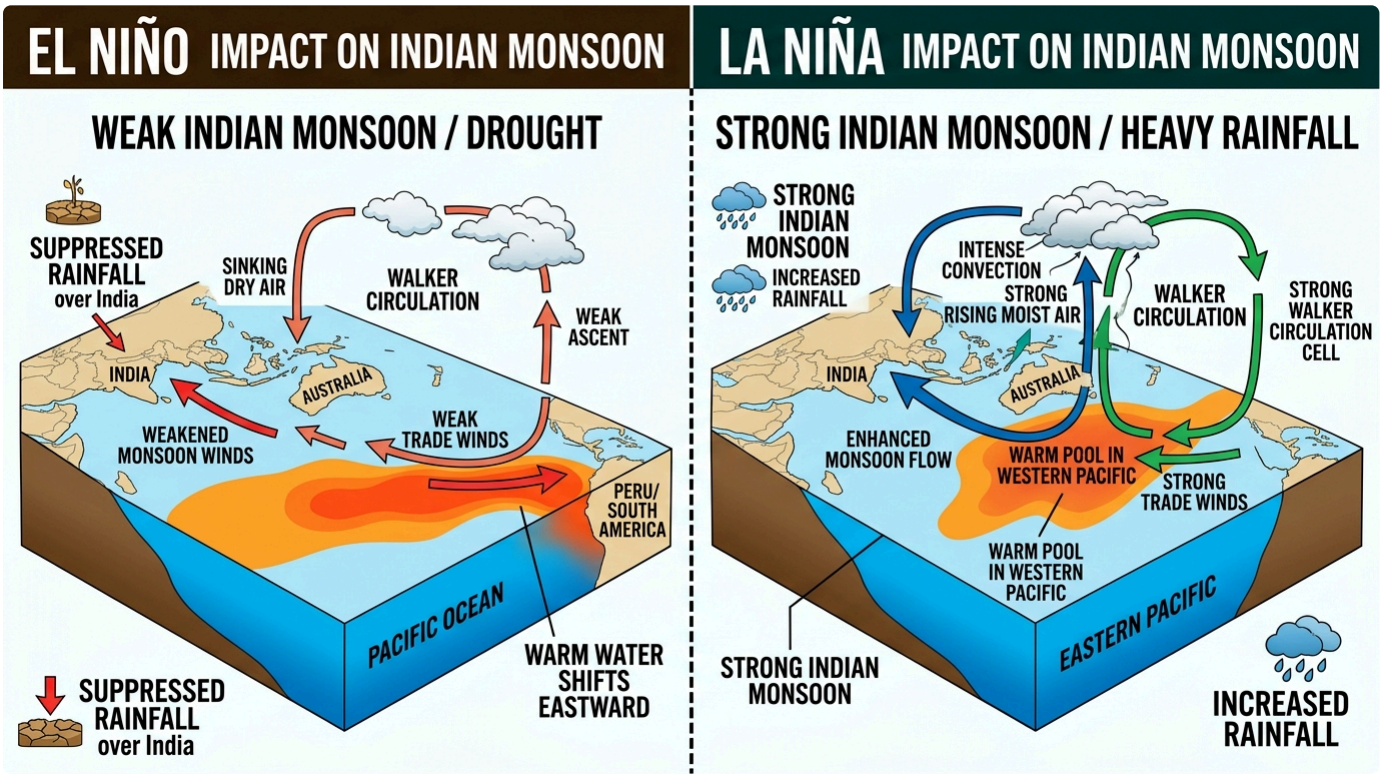
C. Both true, R not explanation

D. (A) true, (R) false

Show answer

Ans: B — **(A) false:** general mid-latitude jets are **westerly**; TEJ is only the summer Indian exception. **(R) true:** 300–500 km/h is the usual class figure, but speed does not save A. **A/C** wrongly accept “easterly.” **D** rejects a true speed line.

El Niño, La Niña, and IOD



These three names scare many students. Keep them separate first.

- **El Niño** and **La Niña** happen in the **Pacific Ocean** (near Peru / South America).
- **IOD** happens in the **Indian Ocean**.
- All three can change India’s monsoon, but they are **not** the same thing.

First understand a normal Pacific year

Look at the Pacific from Peru (east) to Indonesia / Australia (west).

In a normal year:

1. Trade winds blow from **east to west** across the Pacific.

2. They push warm surface water toward the **west** (Indonesia side).
3. Near **Peru** (east), cold deep water rises to the surface. This rise is called **upwelling**.
4. Cold upwelling water brings nutrients. **Plankton** grow. Fish that eat plankton also do well.
5. Rain prefers the warm western Pacific.

So normal year fact: **Peru cool + upwelling on + plankton good**. West Pacific warmer and rainier.

El Niño

Meaning: the eastern Pacific near Peru becomes unusually **warm**.

What happens step by step:

1. Trade winds become **weak**.
2. Warm water stays near Peru instead of being pushed fully west.
3. Cold upwelling near Peru **stops or weakens**.
4. Nutrients fall. So **plankton fall** and fish catch falls. trap: warm water does **not** increase plankton.
5. Rain shifts toward the central / eastern Pacific.
6. For India: monsoon circulation often becomes weaker. So El Niño years **tend** to give a **weaker** Indian monsoon / drought risk. This is a tendency, not a fixed law.

El Niño is a warm current phase off **Peru**. Weak upwelling cuts nutrients, so plankton and fish fall. India's monsoon often weakens in El Niño years.

La Niña

Meaning: the opposite of El Niño.

What happens:

1. Eastern Pacific near Peru becomes **extra cool**.
2. Trade winds become **stronger**.
3. Upwelling near Peru is strong.
4. For India: monsoon often becomes **stronger** / wetter.

La Niña cools the Peru side of the Pacific. India's monsoon often strengthens in La Niña years.

What is ENSO then?

ENSO means **El Niño–Southern Oscillation**. It joins the ocean temperature change with the linked pressure swing.

- El Niño and La Niña are the **ocean** temperature phases.
- The Southern Oscillation is the linked **pressure** change between Darwin (Australia side) and Tahiti (Pacific side).
- Ocean phase plus pressure swing together make ENSO.

You do not need two separate stories. If a question says ENSO, think El Niño / La Niña family in the Pacific.

IOD (Indian Ocean Dipole)

IOD is **only** about the **Indian Ocean**. Do not mix it with the Pacific.

Compare two sides of the Indian Ocean:

- western Indian Ocean (near Africa / Arabian Sea side)
- eastern Indian Ocean (near Indonesia / Sumatra side)

Positive IOD (+IOD)

- Western Indian Ocean is **warmer** than the eastern Indian Ocean.
- More moisture is available toward India.
- Indian monsoon is usually **helped**.
- A +IOD can even **save** a year that also has El Niño. Example: **2019** had El Niño but +IOD, and monsoon finished near normal.

Negative IOD (–IOD (ऋणात्मक IOD))

- Eastern Indian Ocean is **warmer**.
- Indian monsoon is usually **hurt / suppressed**.
- If El Niño and –IOD come together, drought risk rises.

UPSC trap: IOD is west Indian Ocean versus **east Indian Ocean**. It is **not** west Indian Ocean versus eastern **Pacific**.

How to use them in questions

Name	Ocean	India monsoon	Note
El Niño	Pacific (warm Peru)	Often weaker	Plankton **decrease**
La Niña	Pacific (cool Peru)	Often stronger	Opposite of El Niño
+IOD	Indian Ocean (west warmer)	Helps	Can offset El Niño
–IOD	Indian Ocean (east warmer)	Hurts	Can worsen drought

Final rule: never say “El Niño always means drought in India.” Say “El Niño **tends** to weaken the monsoon,” and check whether IOD is helping or hurting in that year.

PYQ — UPPCS Prelims 2023, Q67Logic: Stmt 1 is the classic warm-Peru definition. Stmt 2 wrongly says plankton increase.

With reference to El Niño, which of the following statements is/are correct?

1. El Niño involves the appearance of a warm current off the coast of Peru in the eastern Pacific.
2. This warm current increases the temperature of water on the Peruvian coast by about 10°C, thereby increasing the amount of plankton in the sea.

A. Only 1

B. Only 2

C. Both 1 and 2

D. Neither 1 nor 2

Show answer

Ans: A — Stmt 1 true: warm current off Peru. Stmt 2 false: warm El Niño water weakens upwelling, so plankton and fish **fall**, they do not rise. **C** is the classic trap.

